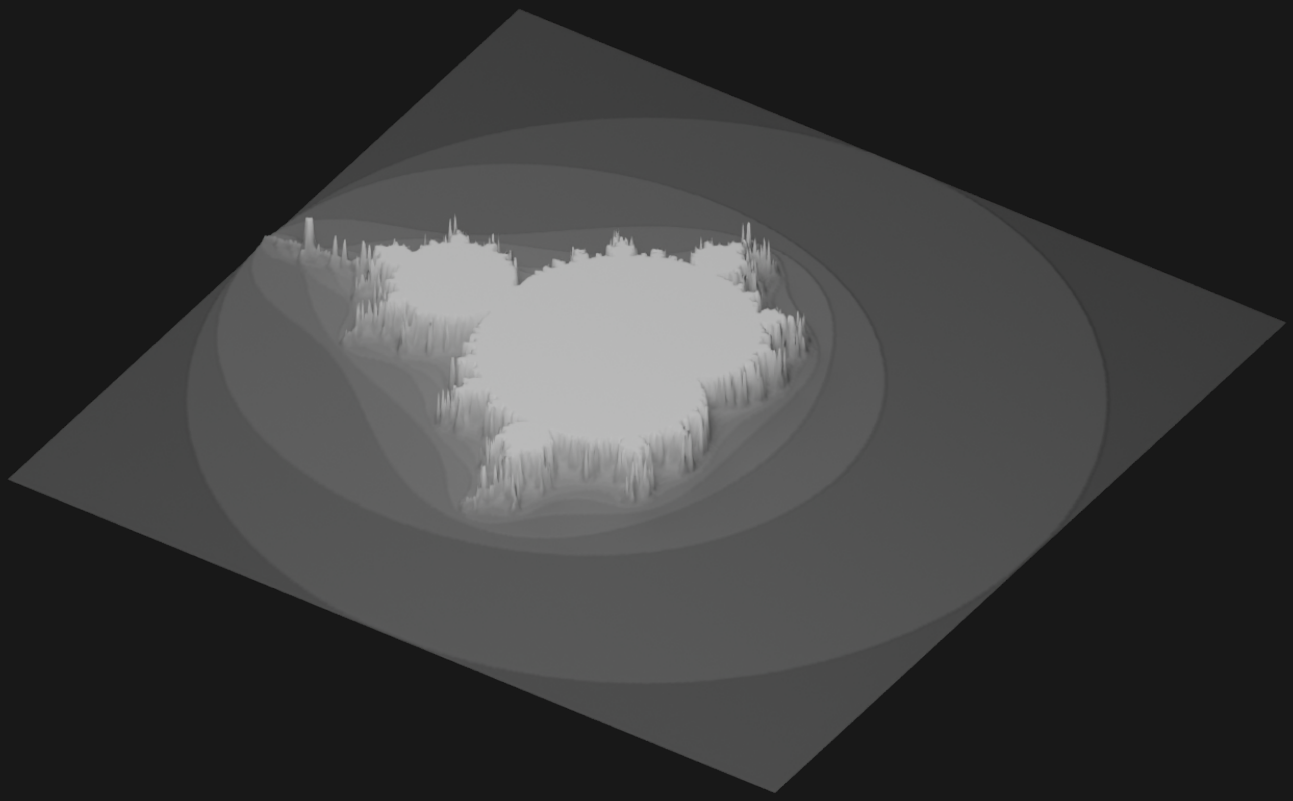
TECHNICAL PORTFOLIO

FRACTALS

From recurrence to geometry

Implemented in Blender Geometry Nodes, a visual programming language, using recursive construction, probabilistic maps and complex iteration.



Mandelbrot escape-time surface. Height and grayscale both encode iteration count.
Actual generated geometry, shown with neutral presentation lighting.

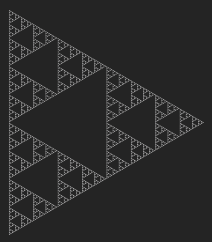
10	3	2D / 3D
FRACTAL GENERATORS	**COMPUTATIONAL APPROACHES**	**GEOMETRIC OUTPUTS**

Complex arithmetic, affine maps and spatial construction in one procedural system

SYSTEM OVERVIEW

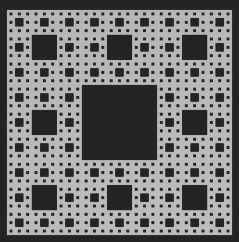
One interface, ten generators

In this visual language, each generator is a node group. A menu switch selects its output, while modifier panels expose the relevant parameters.



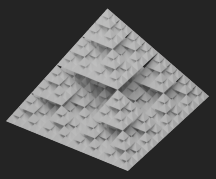
Sierpinski Triangle

3 copies at half scale.
Shown: k = 7.



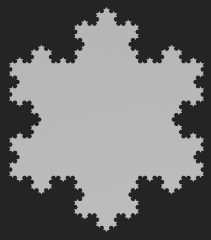
Sierpinski Carpet

8 copies at one-third scale.
Shown: k = 4.



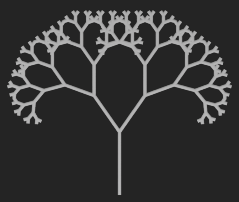
Sierpinski Pyramid

Half-scale copies in 3D.
Square-base mode; k = 4.



Koch Snowflake

4 transformed thirds per side.
Filled mode; k = 4.



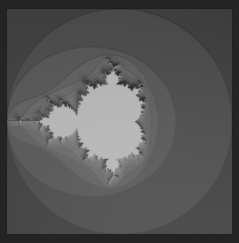
Fractal Tree

Two subtrees plus a trunk.
k = 7; angle = 35 degrees.
Tube thickness control = 3.



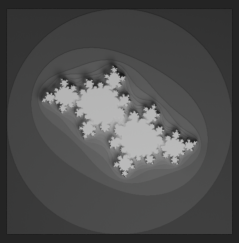
Barnsley Fern

Probabilistic affine maps.
100,000 points; k = 20.



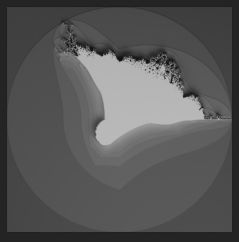
Mandelbrot

Vary c; start from z = 0.
500 x 500 samples; K = 30.



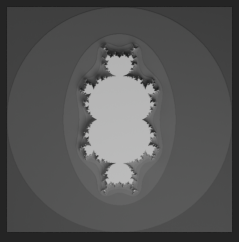
Julia

Vary z0; fix c = -0.4 + 0.6i.
500 x 500 samples; K = 30.



Burning Ship

Fold components before squaring.
500 x 500 samples; K = 30.



Cubic Multibrot

Cube z in polar form.
500 x 500 samples; K = 30.

These are outputs of the same node system. Capture settings are stated above; presentation lighting and grayscale do not change the recurrence or the original modifier configuration.

01 / GEOMETRIC RECURSION

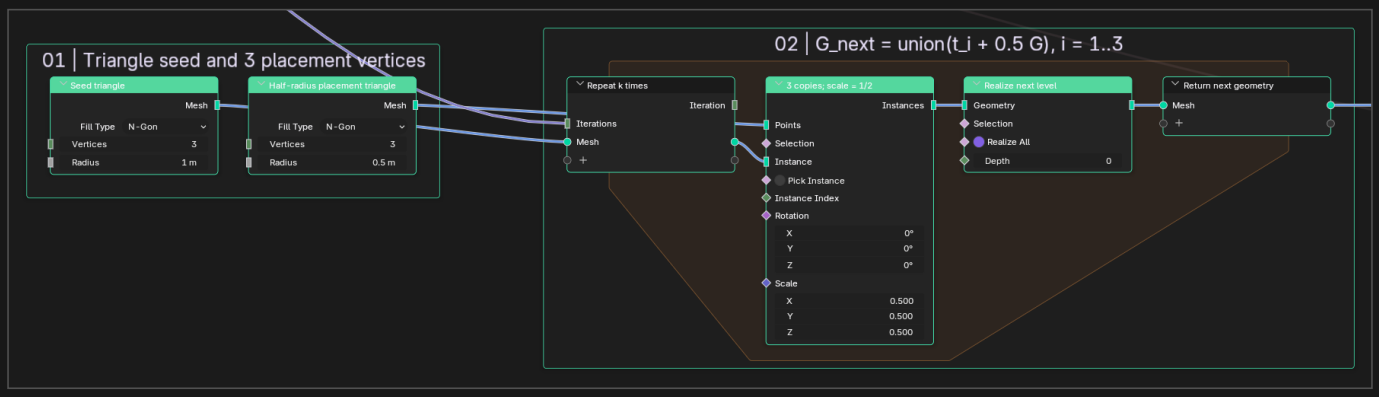
Copies, contraction and topology

A repeat zone replaces the previous geometry with smaller translated copies.
The placement pattern determines which gaps remain at every level.

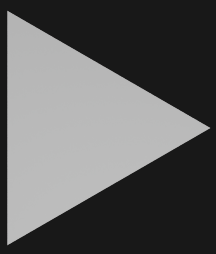
1.1 Triangle: one rule, increasing detail

$$G_{k+1} = \cup_i [t_i + 0.5\, G_k],\ i = 1, 2, 3$$

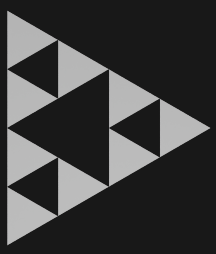
A full-radius triangle supplies the seed. Three vertices at half radius provide the translations. Realize Instances makes the next level available to the repeat zone.



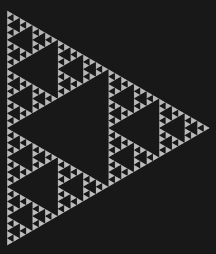
The instance scale is 1/2 on all axes. Merging and final display scaling follow the loop.



k = 0 | 1 triangle



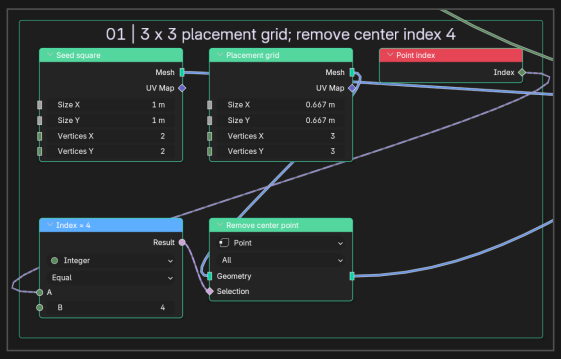
k = 2 | 9 triangles



k = 5 | 243 triangles

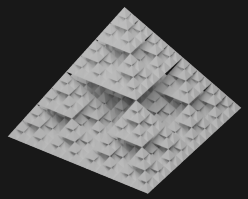
1.2 Carpet: remove the center

A 3 x 3 grid loses point index 4.
Eight retained points place copies at scale 1/3. Face count grows as 8^k.



1.3 Extending the rule into 3D

The pyramid group constructs its seed by extruding a base, collapsing the top to an apex and closing the base. Half-scale copies are placed at the seed vertices.



Square-base seed: 5 copies.
Triangular seed: 4 copies.
Optional reflection creates the paired forms.

01 / GEOMETRIC RECURSION

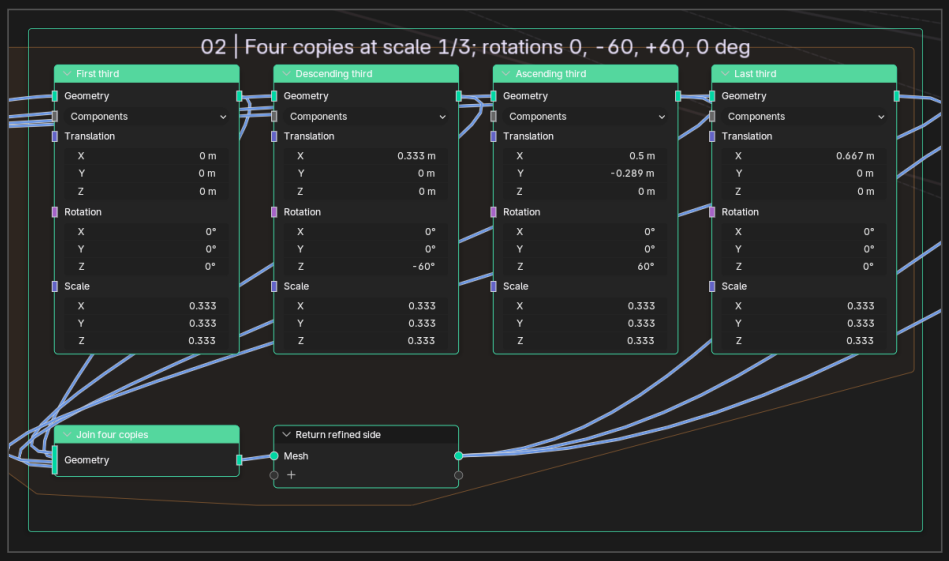
Geometry as a transformation rule

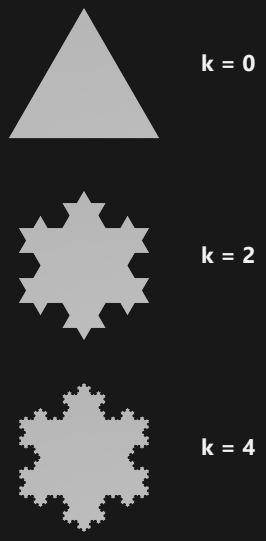
Koch refinement and tree branching use explicit rotations, translations and scale.
The important work is preserving alignment while the structure grows.

1.4 Koch Snowflake

$$G_{k+1} = \cup_i [t_i + (1/3)\, R_i\, G_k]$$

Four transformed thirds replace one side. Their rotations are 0, -60, +60 and 0 degrees. Three copies of the completed side close the snowflake.

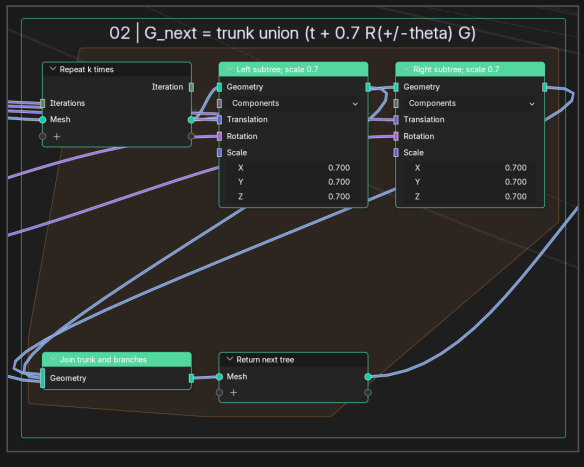




The offsets meet at common endpoints. Merge by Distance joins the assembled sides.
Each refined side has 4^k segments; filled and tubular outputs are presentation options.

1.5 Fractal Tree

$$G_{k+1} = \text{trunk} \cup [t + 0.7\, R(+\theta)\, G_k] \cup [t + 0.7\, R(-\theta)\, G_k]$$



The previous tree is scaled by 0.7, rotated in both directions and moved to the trunk tip. A fresh trunk is joined to the two subtrees at each iteration.

The angle is converted from degrees to radians. A final 90-degree rotation makes the tree upright. The branching occurs in a plane; tube thickness adds volume.



02 / PROBABILISTIC AFFINE SYSTEMS

A fern from four local maps

The fern uses many independent points. Each iteration applies one affine map to each point, chosen by a reproducible pseudorandom value.

2.1 Transform selection

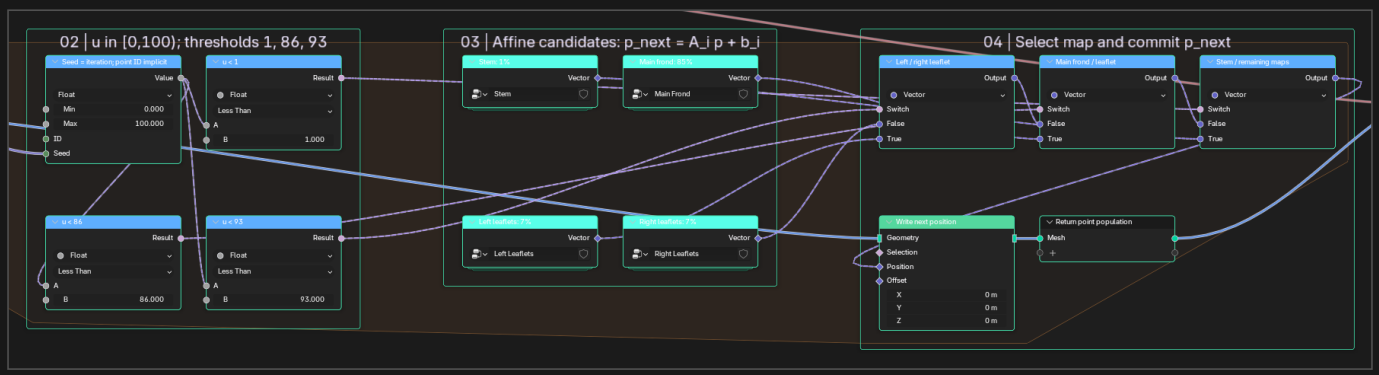
$$p' = A_i\, p + b_i$$



The random seed changes with the iteration.
The implicit point ID distinguishes the samples.
Thresholds at 1, 86 and 93 split [0,100) into probabilities of 1%, 85%, 7% and 7%.

2.2 Coefficients in this implementation

1%	**Stem**	$x' = 0$ $y' = 0.16y$
85%	**Main frond**	$x' = 0.85x + 0.04y$ $y' = -0.04x + 0.85y + 1.6$
7%	**Left leaflets**	$x' = 0.20x - 0.26y$ $y' = 0.23x + 0.22y + 1.6$
7%	**Right leaflets**	$x' = -0.15x + 0.28y$ $y' = 0.26x + 0.24y + 0.44$



Nested switches select a candidate vector. Set Position commits the chosen map.
The repeat carries the point population forward; it does not accumulate every intermediate point.

$$x' = 0.85x + 0.04y$$

$$y' = -0.04x + 0.85y + 1.6$$

MAIN-FROND MAP

The main-frond helper expands its affine map into scalar products, sums and translation. All points begin at the origin. More iterations increase transformation depth; Point Count controls the number of final samples.

03 / COMPLEX ITERATION

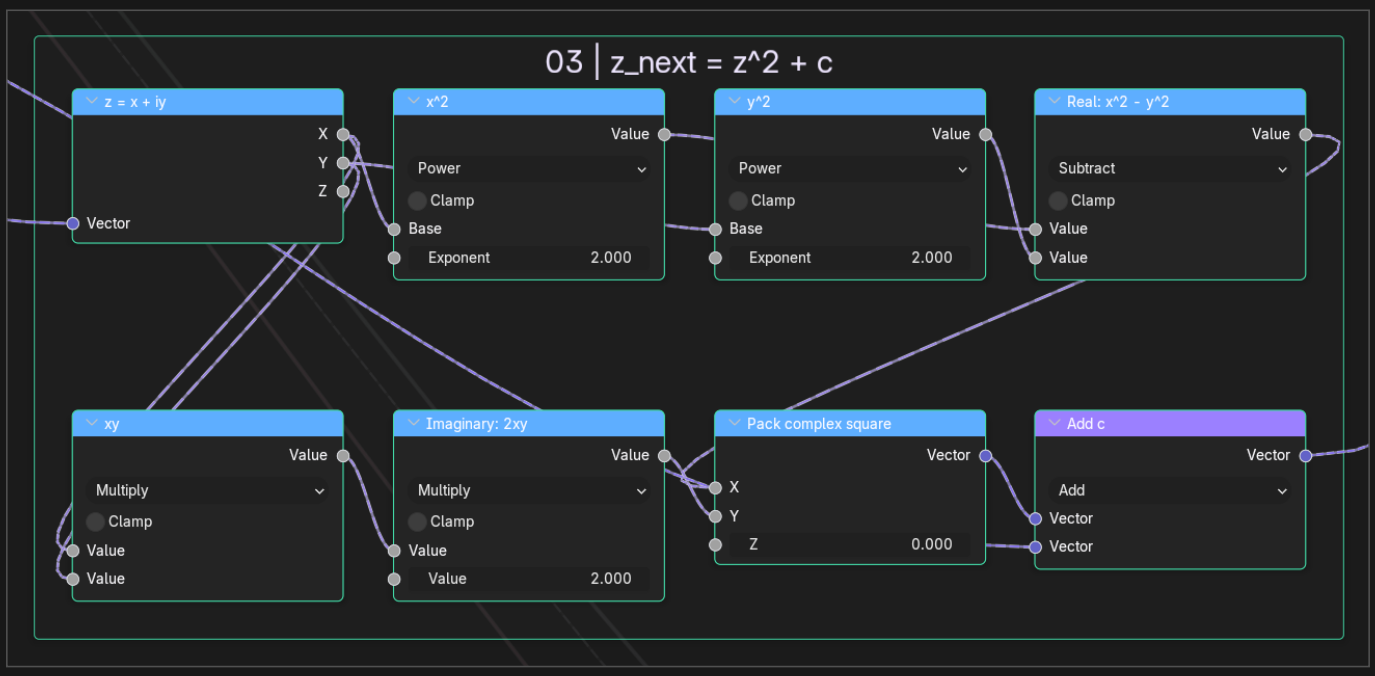
A recurrence evaluated over a grid

The Mandelbrot group treats each grid position as a complex parameter c.
It stores the recurrence state explicitly in point-domain attributes.

3.1 Expand the complex square

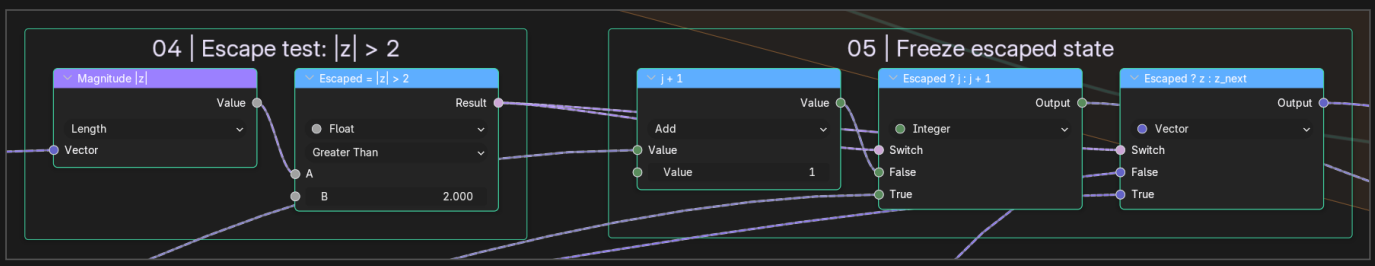
$$z = x + iy \quad z' = z^2 + c \quad z_0 = 0$$

$$x' = x^2 - y^2 + c_x \quad y' = 2xy + c_y$$



Two squares and a subtraction form the real component. Two products form the imaginary component. Combine XYZ packs the pair; vector addition applies c.

3.2 Escape, freeze and count



State: C holds the parameter, Z the current complex value, and Iter the counter j.
If |z| > 2, switches retain z and j. Otherwise the next state is written and j increases.

Output: h = H j / K displaces the sample grid vertically. The check uses the current z, before the update. The loop still runs K times; escaped samples remain frozen.

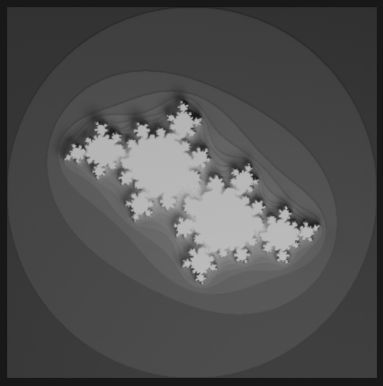
At R = 500 and K = 30, the grid has 250,000 vertices. Samples reaching the cap have h = H.

03 / COMPLEX ITERATION

Change the map, change the structure

The same state-and-escape pattern supports three variants.
Changing initialization or one recurrence changes the resulting geometry.

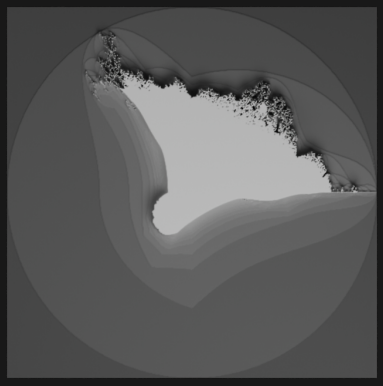
Julia



$z' = z^2 + c$

Grid position sets z0.
The exposed constant is
c = -0.4 + 0.6i.

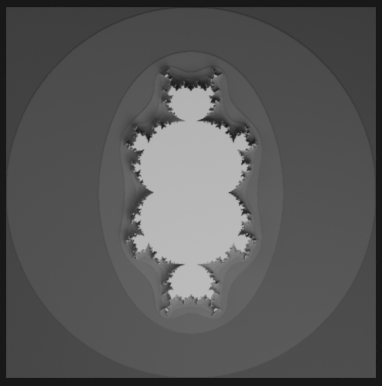
Burning Ship



$z' = (|x| + i|y|)^2 + c$

Absolute values fold the plane
before the quadratic step.
A final rotation sets the display.

Cubic Multibrot



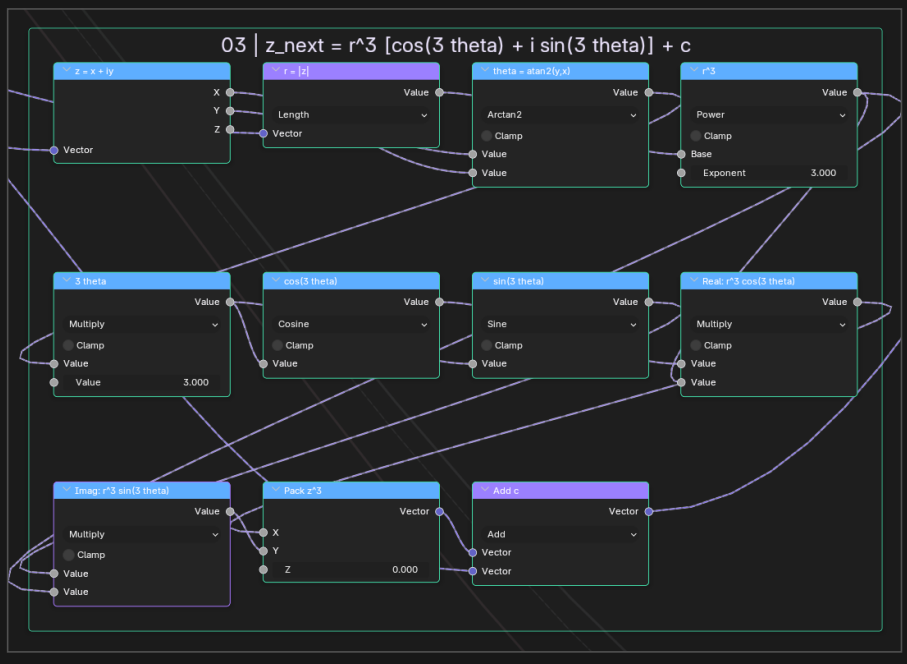
$z' = z^3 + c$

The implementation fixes d = 3.
Both radius and angle are
transformed before adding c.

3.3 Cubing through polar coordinates

$r = \sqrt{(x^2 + y^2)} \quad \theta = \text{atan2}(y, x)$

$z^3 = r^3 [\cos(3\,\theta) + i \sin(3\,\theta)]$



The Length node gives r.
Arctan2 preserves the
quadrant of the angle.

The graph computes r^3
and 3 theta separately,
then reconstructs the
real and imaginary parts.

This is a geometric
interpretation of complex
multiplication, expressed
as node operations.

COMPUTATION AND SCOPE

What the system demonstrates

The project implements established mathematical constructions as inspectable procedural graphs, with controls for depth, sampling and presentation.

4.1 Two different kinds of computational growth

Geometric replication grows exponentially with depth: 3^k triangle copies, 8^k carpet copies and 4^k segments per Koch side. Realizing geometry makes that growth explicit in memory. The fern keeps P samples and repeats K maps.

ESCAPE-TIME GRID: NOMINAL R^2 K SAMPLE-ITERATIONS

Resolution R	Vertices R^2	At K = 30
250	62,500	1.875 million
500	250,000	7.5 million
1,000	1,000,000	30 million

4.2 What the output means

Finite depth approximates self-similar structure. For escape-time sets, remaining bounded for K steps is a finite observation, not proof of permanent membership.

Height encodes a counter. Tube thickness, materials and grayscale help reveal the result; they are not part of the mathematical definition.

4.3 Technical capability

Complex arithmetic becomes scalar operations. Affine maps connect linear algebra to shape. Transformation rules require consistent spatial alignment and awareness of scale.

Blender Geometry Nodes is the visual language used to express these algorithms. The project contains no learned model or physics simulation.

4.4 Evidence and possible extensions

Actual outputs were captured for all ten generators. Triangle examples at k = 0, 2 and 5 contain 1, 9 and 243 faces, matching 3^k. The inspected Mandelbrot grid has 250,000 vertices and 249,001 faces. State values were inspected directly.

Useful next steps would be exposing the Multibrot exponent, adding a complex-plane zoom transform, and comparing resolution against iteration depth. These are proposed extensions; runtime performance and numerical error have not been benchmarked.

Source and implementation credits

Source: the accompanying Blender file and its named fractal groups. Equations and coefficients describe the connected implementation. Curve to Tube and its dependencies are linked Blender Essentials assets used for presentation. All figures show actual outputs or node-editor captures; the grayscale surface shading encodes the stored iteration count.